

Harvard Medical School Curriculum Vitae

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Education

04/1997	Vordiplom, BS	Physics, mathematics, chemistry	Albert-Ludwigs-Universität Freiburg, Germany
12/2000	Diplom, MS	Physics	Albert-Ludwigs-Universität Freiburg, Germany
08/2005	MS	Physics	Vanderbilt University Nashville, TN
08/2006	PhD	Physics Advisor: F J Baudenbacher	Vanderbilt University Nashville, TN
05/2017	Professional Certificate	Data Science	Harvard University Extension School

Postdoctoral Training

06/06 - 05/09	Postdoctoral Fellow	Cardiovascular Research Center (CVRC) PI: Calum A MacRae	Massachusetts General Hospital (MGH) Harvard Medical School (HMS)
08/09 - 06/12	Postdoctoral Fellow	Heart and Vascular Center Cardiac Imaging Lab PI: David S Rosenbaum	MetroHealth Hospital Case Western Reserve University (CWRU), Cleveland, OH

Faculty Academic Appointments

2012 - 2015	Instructor	Cardiovascular Division Medicine	HMS
2015 - 2022	Assistant Professor	Cardiovascular Division Medicine	HMS
2026 -	Research Scientist	Biomedical Informatics	HMS

Appointments at Hospitals/Affiliated Institutions

2001 – 2006	Research Assistant	Physics	Vanderbilt University Nashville, TN
2012 -	Associate Scientist	Cardiovascular Division	Brigham and Women's Hospital Boston, MA

2022 -	Research Associate	Department of Biomedical Informatics (DBMI)	HMS
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Other Professional Positions

2016 - 2017	Consultant		TTP plc, Melbourn, UK
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Major Administrative Leadership Positions

Local

2024 -	Director of AI and Data Science	Core for Computational Biomedicine	DBMI, HMS
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Committee Service

Local

2011 - 2012	Dissertation Advisory Committee Michelle Jennings, PI: David S Rosenbaum		Case Western Reserve University
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2026	Faculty for oral radiology search committee		Harvard School of Dental Medicine
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Professional Societies

1994 -	German Physical Society (DPG)	Member
2002 -	American Physical Society (APS)	Member
2005 -	Biomedical Engineering Society (BMES)	Member
2011 -	American Heart Association (AHA)	Member

Editorial Activities

Ad hoc Reviewer

Science

PNAS

Journal of Physiology, London

Progress in Biophysics and Molecular Biology

Pacing and Clinical Electrophysiology

Journal of Clinical Investigation

Honors and Prizes

2001	Scholarship	VU	Research
2006	Multidisciplinary Training in Cardiovascular Research T32HL007208	MGH	Research
2008	Best Abstract CVRC	MGH	Research
2010	Best Abstract CWRU	CWRU	Research
2013	Travel Award	Gordon Research Conferences	Research
2016	Cardiovascular Division Chair's Research Award	BWH	Research

Report of Funded and Unfunded Projects

Past

- 2024 - 2025 A Framework for Simulation-Based Dental Education Using Generative Modeling
HMS Dean's Innovation Awards for the Use of Artificial Intelligence in Education, Research, and Administration
Role: Co-Investigator PI: Magda Feres, DDS, PhD
The goal of this project was to develop a framework for simulation-based learning through generative AI.
- 2010 - 2012 Molecular Mechanisms of stretch-induced Electrical Remodeling in the Heart
American Heart Association (AHA) Postdoctoral Fellowship Award
AHA 10POST4150110
Role: PI (\$88,000)
The goal of this project was to establish and validate a zebrafish model of stretch-induced Ventricular Electrical Remodeling (VER). This model is used to study the detrimental electrophysiological consequences of short-term and long-term mechanical strain, which is known to cause cardiac arrhythmias and sudden death in patients with conduction system defects and ventricular pacing.
- 2012 Development of an in vitro arrhythmia susceptibility assay using human-induced, pluripotent stem (iPS) cells. Cellular Dynamics International Innovative Research Award
Role: PI (\$100,000)
This project aimed to demonstrate that cardiac arrhythmia mechanisms can be identified in engineered cardiac tissues from human-derived iPS cells.
- 2012 - 2019 Molecular Mechanisms of stretch-induced Electrical Remodeling in the Heart
NIH NHLBI Pathway to Independence Award 5K99HL112724
Role for K99 Phase: PI (\$264,600). Project Period: 08/21/2012 - 11/30/2014
Role for R00 Phase: PI (\$750,000). Project Period: 06/01/2015 - 06/30/2019
The goal of this project was to understand the molecular mechanisms of stretch-induced ventricular electrical remodeling (VER) using the zebrafish as an animal model. Results from this study will lead to new therapeutic strategies to treat VER and cardiac arrhythmias in patients with conduction system defects or ventricular pacing.

Unfunded Current Projects

- 2023 - Estimating Periodontal Stability Using Computer Vision
PI: Magda Feres, DDS, PhD
Role: Project team lead
This project aims to create a state-of-the-art computer vision model to estimate periodontal stability from dental radiographs.
- 2024 - Authentically integrating artificial intelligence into assessments and tutoring within a graduate-level molecular biology course at Harvard Medical School
PI: Joseph Loparo, PhD
Role: Project team lead
This project aims to create an interactive learning platform that lets students interact with large language models to ask questions about recorded lectures and course materials.

- 2024 - From ARI to AIRI: Examining Generative AI Applied within — and with — Research Integrity
 PI: Blake Talbot, MPH
 Role: Project team lead
 This project aims to develop generative AI solutions that help the Office for Academic and Research Integrity (ARI) facilitate its research misconduct reviews and build a knowledge base for the appropriate evaluation of allegations of research misconduct involving the use (and potential misuse) of AI by a researcher.
- 2024 - An AI-powered syllabus app to support course design and teaching practice
 PI: Tari Tan, PhD
 Role: Project team lead
 This project aims to develop an AI-powered app that generates and evaluates course syllabi against established rubrics, giving students and faculty immediate design feedback to support curriculum innovation, efficiency, and professional development. After refinement and integration into the HMS Canvas portal, it will streamline course planning and enhance teaching practice in medical education.

Report of Local Teaching and Training

Formal Teaching of Residents, Clinical Fellows, and Research Fellows (post-docs)

2009 - 2012	Research update in cardiac electrophysiology Graduate students, post-docs	MetroHealth, CWRU 1 lecture per month
2012 - 2015	Cardiovascular research update Cardiology fellows, post-docs, lab technicians	BWH 1 lecture per month
2021	Electrophysiology journal club	BWH 1 lecture per month
2024	Research update for CCB	HMS 4 lectures per year

Research Supervisory and Training Responsibilities

2011 - 2012	Supervision of undergraduate summer students	CWRU Daily mentorship 3 months per year
2012 - 2018	Supervision of post-doctoral research fellows (average 1-2 fellows per year)	BWH Daily mentorship
2022 -	Supervision of post-doctoral fellows and team members (average 1-2 fellows per year)	DBMI, CCB One hour meeting per week; 1:1 supervision one hour per week per fellow

Formally Mentored Harvard Students (Medical, Dental, Graduate, and Undergraduate)

2024 - 2025	Krishna Patel, student, Harvard University <i>Career stage:</i> undergraduate student. <i>Mentoring role:</i> research advisor Project title: Artificial intelligence to integrate multi-day images for enhanced detection of surgical site infections in rural Rwanda	
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Other Mentored Trainees and Faculty

2011 - 2012	Anna Brzezinski, student, University of Pennsylvania, PA <i>Career stage:</i> undergraduate student. <i>Mentoring role:</i> research advisor <i>Accomplishments:</i> 1 manuscript published (research investigation 19)
2012 - 2015	Michelle Nassal, PhD candidate, CWRU <i>Career stage:</i> graduate student. <i>Mentoring role:</i> research advisor <i>Accomplishments:</i> 1 manuscript published (research investigation 25) Student received an NIH T32 predoctoral fellowship award
2013 - 2015	Arndt Anne-Karin, MD, BWH <i>Career stage:</i> post-doctoral fellow. <i>Mentoring role:</i> research advisor <i>Accomplishments:</i> 1 manuscript published (research investigation 21)
2014 - 2017	Brittney M Mikell, laboratory technician, BWH <i>Career stage:</i> pre-doctoral student. <i>Mentoring role:</i> research advisor <i>Accomplishments:</i> abstract presented at the AHA annual meeting
2014 - 2015	David White, PhD, BWH <i>Career stage:</i> post-doctoral fellow. <i>Mentoring role:</i> research advisor
2015 - 2017	Sunil Kapur, MD, BWH <i>Career stage:</i> post-doctoral fellow. <i>Mentoring role:</i> research advisor <i>Accomplishments:</i> 2 manuscripts (research investigations 28, 31)
2015 - 2016	Man Qi, MD, BWH <i>Career stage:</i> post-doctoral fellow. <i>Mentoring role:</i> research advisor
2018 - 2019	David Plotnick, BWH <i>Career stage:</i> pre-doctoral student. <i>Mentoring role:</i> research advisor <i>Accomplishments:</i> 1 manuscript published (research investigations 31)
2020 - 2021	Zara Adamou, student, Weill Cornell Medicine, NY <i>Career stage:</i> student. <i>Mentoring role:</i> research advisor
2024 - 2025	Sreya Banerjee, PhD, DBMI, HMS <i>Career stage:</i> post-doctoral fellow. <i>Mentoring role:</i> research advisor <i>Accomplishments:</i> Faculty position in AI/ML at SUNY New Paltz, NY
2025 - 2026	Nidia I Bucarelli, PhD <i>Career stage:</i> post-doctoral fellow. <i>Mentoring role:</i> supervisor <i>Accomplishments:</i> 1 manuscript in preparation

Local Invited Presentations

No presentations below were sponsored by 3rd parties/outside entities

2011	Cardiac electrical remodeling triggered by mechanical stretch. New evidence from the zebrafish model MetroHealth Heart and Vascular Research, CWRU
2015	Seminars in Cardiology The zebrafish as a model to study connectivity in the heart Beth Israel Deaconess Medical Center, Boston, MA

2023	Global symposium on artificial intelligence in dentistry Harvard School of Dental Medicine, Boston, MA
2024	Introduction to version control with Git and GitHub Health Care Policy, Blavatnik Institute, HMS
2025	2nd global symposium on artificial intelligence in dentistry Harvard School of Dental Medicine, Boston, MA
2026	Augmented diagnostics and computer vision HMS / HSDM Boston, MA

Report of Regional, National and International Invited Teaching and Presentations

Invited Presentations and Courses

Those presentations below sponsored by 3rd parties/outside entities are so noted and the sponsors are identified.

Regional

2014	A developmental cardiac connectome in the zebrafish reveals complex myocyte circuits/ Invited lecture University of Massachusetts Medical School/Department of Cardiovascular Medicine Worcester, MA
2025	Applied computer vision in dental medicine / invited presentation Osteology Research Academy 2025 Boston, MA

National

2013	Electrophysiological heterogeneity in the developing heart: Prerequisite for life or cause of sudden death? Invited lecture at the Cardiac Electrophysiology Symposium honoring Dr. David Rosenbaum at MetroHealth Medical Center, Cleveland, OH
2015	Cellular connectivity in the developing embryo: what can we learn from the zebrafish model?/Invited lecture Society of Photo-Optical Instrumentation Engineers (SPIE) conference San Francisco, CA
2019	Stretch and cardiac electrical remodeling/Invited lecture American Heart Association Scientific Sessions Arrhythmia Research Summit, Philadelphia, PA

International

2012	Cardiac Electrical Remodeling Triggered by Stretch: What can we learn from the zebrafish? /Invited lecture Max Delbrück Center for Molecular Medicine Berlin, Germany
2015	Innovative approaches to drug discovery and safety screening/Invited Presentation The Technology Partnership, TTP Melbourn, UK

Report of Technological and Other Scientific Innovations

2001 Instrument Ultra-High Vacuum Laser Ion Source	As a graduate student at the Albert-Ludwigs Universität Freiburg, I designed, built, and tested an ultra-high vacuum laser ion source to inject lowly charged ions into an electron beam ion trap. The device was used at the Max-Planck Institute for Nuclear Physics in Heidelberg, Germany, to study the interactions of highly charged ions. Research investigations 1, 2, 3
2002 Patent Cell-based biosensors	As a graduate student at Vanderbilt University, I developed cell-based biosensors for the Biosensor Technologies Program of the Defense Advanced Research Projects Agency (DARPA). The technology uses cells living on a microfluidic chip to identify known and unknown biological warfare agents. US Patent Application, 60310652, filed Aug 6, 2001 (http://appft.uspto.gov/) Apparatus for monitoring the status of a metabolically active cell US Patent Application, 10212918, filed Aug 6, 2002 (http://appft.uspto.gov/) Apparatus and methods for using biological material to discriminate an agent
2009 Software	As a research fellow at MGH, I developed a unique software package for analyzing high-speed, high-resolution fluorescence data. Results obtained with this software were validated and published multiple times in peer-reviewed journals (research investigations 12-14, 16, 17, 19, 21). The software is being used at HMS and other laboratories in the US and abroad.
2024 Software	As a research associate at DBMI, I have created an open-source repository for state-of-the-art computer vision. The repository can be used as a template for image classification, object detection, and segmentation models https://github.com/ccb-hms/computervision

Report of Education of Patients and Service to the Community

No activities or materials below were sponsored by 3rd parties/outside entities

Educational Material for Patients and the Lay Community

2013 - 2015	Boston Latin School, Boston, MA First laboratory experience for high school students
2016	A Strategic Vision for 21st-century Drug Discovery Bowen W, Sanders G, Werdich AA , Warrington B, Farrant E Drug Discovery World Winter 2016/17 www.ddw-online.com/a-strategic-vision-for-21st-century-drug-discovery-594-201612/
2021 - 2022	COVID, Activity Patterns, Physiology, and Stress Study Insights and Education Apple Heart and Movement Study https://appleheartandmovementstudy.bwh.harvard.edu/category/Study-Updates/

Report of Scholarship ORCID ID: 0000-0002-1222-1180

Peer-reviewed publications in print or other media

Research Investigations

1. Lopez-Urrutia JRC, Bapat B, Feuerstein B, **Werdich AA**, Ullrich J
First results from the Freiburg electron beam ion trap FreEBIT
Hyperfine interactions 2000; 127(4):497-501

2. Lopez-Urrutia JRC, Bapat B, Draganic I, **Werdich AA**, Ullrich J
First results from the Freiburg electron beam ion trap FreEBIT
Physica scripta 2001; T92:110-112
3. Mironov V, Trinczek M, **Werdich AA**, Martinez AJG, Guo P, Zhang X, Braun J, Lopez-Urrutia JRC, Ullrich J
Tests of a laser ion source at the Heidelberg electron beam ion trap
Nuclear instruments & methods in physics section B 2003; 205:183-86
4. Baudenbacher F, Monroe TW, **Werdich AA**, Cliffler D, Wikswo JP
The nanophysiometer: BioMEMS for high bandwidth detection of cellular activity in subnanoliter volumes
Proceedings of the Second Joint EMBS/BMES Conference, Houston, TX 2002
5. **Werdich AA**, Lima EA, Ivanov BL, Ges IA, Anderson ME, Wikswo JP, Baudenbacher FJ
A microfluidic device to confine a single cardiac myocyte in a sub-nanoliter volume on planar microelectrodes for extracellular potential recordings
Lab on a chip 2004; 4(4):357-362
6. Ges IA, Ivanov BL, Schaffer DK, Lima EA, **Werdich AA**, Baudenbacher FJ
Thin-film IrOx pH microelectrode for microfluidic-based microsystems
Biosensors & bioelectronics 2005; 21(2):248-256
7. Trinczek M, **Werdich AA**, Mironov V, Guo P, González Martínez AJ, Braun J, Crespo López-Urrutia JR, Ullrich J. A laser ion source for an electron beam ion trap
Nuclear instruments & methods in physics section B 2006; 251(1):289-296
8. Ges IA, Ivanov BL, **Werdich AA**, Baudenbacher FJ
Differential pH measurements of metabolic cellular activity in nl culture volumes using microfabricated iridium oxide electrodes
Biosensors & bioelectronics 2007; 22(7):1303-1310
9. **Werdich AA**, Baudenbacher FJ, Dzhura I, Jeyakumar LH, Kannankeril PJ, Fleischer S, LeGrone A, Milatovic D, Aschner M, Atkinson J, Strauss AW, Anderson ME, Baudenbacher FJ, Exil VJ
Polymorphic ventricular tachycardia and abnormal Ca^{2+} handling in very-long-chain acyl-CoA dehydrogenase null mice
American journal of physiology, heart and circulatory physiology 2007; 292(5): H2202-2211
10. **Werdich AA**, Lima EA, Dzhura I, Singh MV, Li J, Anderson ME, Baudenbacher FJ
Differential effects of phospholamban and Ca^{2+} /calmodulin-dependent kinase II on $[\text{Ca}^{2+}]_i$ transients in cardiac myocytes at physiological stimulation frequencies
American journal of physiology, heart and circulatory physiology 2008; 294(5): H2352-2362
11. Kokel D, Bryan J, Laggner C, White R, Cheung CY, Mateus R, Healey D, Kim S, **Werdich AA**, Haggarty SJ, MacRae CA, Shoichet B, Peterson RT
Rapid behavior-based identification of neuroactive small molecules in the zebrafish
Nature chemical biology 2010; 6(3):231-237 *PMCID* 2834185
12. Kikuchi K, Holdway JE, **Werdich AA**, Anderson RM, Fang Y, Egnaczyk GF, Evans T, MacRae CA, Stainier DY, Poss KD, Primary contribution to zebrafish heart generation by gata4(+) cardiomyocytes
Nature 2010; 464(7288):601-605 *PMCID* 3040215
13. Panakova D*, **Werdich AA***, MacRae CA
*Authors contributed equally to this work
Wnt11 patterns a myocardial electrical gradient through regulation of the L-type Ca^{2+} channel
Nature 2010; 466(7308):874-878 *PMCID* 2921013
14. Becker JR, Deo RC, **Werdich AA**, Panakova D, Coy S, MacRae CA
Human cardiomyopathy mutations induce myocyte hyperplasia and activate hypertrophic pathways during cardiogenesis in zebrafish
Disease models & mechanisms 2011; 4(3):400-410 *PMCID* 3097461

15. Pong T, Adams WJ, Bray MA, Feinberg AW, Sheeny SP, **Werdich AA**, Parker KK
Hierarchical architecture influences calcium dynamics in engineered cardiac muscle
Experimental biology and medicine 2011; 236(3):366-373
16. Wythe JD, Jurynece MJ, Urness LD, Jones CA, Sabeh MK, **Werdich AA**, Sato M, Yost HJ, Grunwald DJ, MacRae CA, Li DY. Hapd1, a newly identified pleckstrin homology domain protein, is required for cardiac contractility in zebrafish
Disease models & mechanisms 2011; 4(5):607-621 *PMCID* 3180224
17. Wang J, Panakova D, Kikuchi K, Hodway JE, Gemberling M, Burrus JS, Singh SP, Dickson AL, Lin YF, Sabeh MK, **Werdich AA**, Yelon D, MacRae CA, Poss KD
The regenerative capacity of zebrafish reverses cardiac failure caused by genetic cardiomyocytes depletion
Development 2011; 138(16):3421-3430 *PMCID* 3143562
18. Feinberg AW, Alford PW, Jin H, Ripplinger CM, **Werdich AA**, Sheeny SP, Grosberg A, Parker KK
Controlling the contractile strength of engineered cardiac muscle by hierarchical tissue architecture
Biomaterials 2012; 33(23):5732-5741
19. **Werdich AA**, Brzezinski A, Jeyaraj D, Sabeh MK, Ficker EF, Wan X, MacRae CA, Rosenbaum DS
The zebrafish as a novel animal model to study the molecular mechanisms of mechano-electrical feedback in the heart
Progress in Biophysics and Molecular Biology 2012; 110(2-3): 154-165 *PMCID* 3663588
20. Nath AK, Roberts LD, Liu Y, Mahon S, Kim S, Ryu JH, **Werdich A**, Januzzi JL, MacRae CA, Rockwook GA, Brenner MB, Gerszten RE, Peterson RT
Chemical and metabolomic screens identify novel biomarkers and antidotes for cyanide exposure
FASEB Journal 2013; 27(5):1928-1938 *PMCID* 3633825
21. Arndt AK, Schafer S, Drenckhahn JD, Sabeh MK, Plovie ER, Caliebe A, Klopocki E, Musso G, **Werdich AA**, Kalwa H, Heinig M, Padera RF, Wassilew K, Bluhm J, Harnack C, Martitz J, Barton PJ, Greutmann M, Berger F, Hubner N, Siebert R, Kramer HH, Cook SA, MacRae CA, Klaassen S
Fine mapping of the 1p36 deletion syndrome identifies mutation of PRDM16 as a cause of cardiomyopathy
American Journal of Human Genetics 2013; 93(1):67-77 *PMCID* 3710750
22. Roder K*, **Werdich AA***, Li W, Liu M, Kim TY, Organ-Darling LE, Moshal KS, Hwang JM, Lu Y, Choi BR, MacRae CA, Koren G. RING Finger Protein RNF207: A novel regulator of cardiac excitation
*Authors contributed equally to this work
Journal of Biological Chemistry 2014; 289(49):33730-33740 *PMCID* 4256309
23. Gu S, Wang YT, Ma P, **Werdich AA**, Rollins AM, Jenkins MW
Mapping conduction velocity of early embryonic hearts with a robust fitting algorithm
Biomedical Optics Express 2015; 6(6):2138-2157 *PMCID* 4473749
24. Mosimann C, Panakova D, **Werdich AA**, Musso G, Burger A, Lawson KL, Carr LA, Nevis KR, Sabeh MK, Zhou Y, Davidson AJ, DiBiase A, Burns CE, Burns CG, MacRae CA, Zon LI
Chamber identity programs drive early functional partitioning in the heart
Nature Communications 2015; 6:8146 *PMCID* 4560818
25. Nassal M, **Werdich AA**, Wan X, Hoshi M, Deschenes I, Rosenbaum DS, Donahue J
Phosphorylation at connexin43 serine-368 is necessary for myocardial conduction during metabolic stress
Journal of Cardiovascular Electrophysiology 2016; 27(1):110-119 *PMCID* 4879887
26. McPheeters MT, Wang YT, **Werdich AA**, Jenkins MW, Laurita KR
An infrared optical pacing system for screening cardiac electrophysiology in human cardiomyocytes
PLoS One 2017; 12(8) *PMCID* 5570338

27. Nikolova-Krstevski V, Wagner S, Yu ZY, Cox CD, Cvetkovska J, Hill AP, Huttner IG, Benson V, **Werdich AA**, MacRae C, Feneley MP, Friedrich O, Martinac B, Fatkin D
Endocardial TRPC-6 channels act as atrial mechanosensors and load-dependent modulators of endocardial/myocardial cross-talk
JACC Basic to Translational Science 2017; 2(5):575-590 PMID 6058914
28. Colombo S, de Sena-Thomas C, George V, **Werdich AA**, Kapur S, MacRae CA, Targoff KL
Nkx genes establish second heart field cardiomyocyte progenitors at the arterial pole and pattern the venous pole through Isl1 repression
Development 2018; 145(3): PMID 5818010
29. Moshal KS, Roder K, Kabakov AY, **Werdich AA**, Chiang DY, Turan NN, Xie A, Kim TY, Cooper LL, Lu Y, Zhong M, Li W, Terentyev D, Choi BR, Karma A, MacRae CA, Koren G
LITAF (lipopolysaccharide-induced tumor necrosis factor) regulates cardiac L-type calcium channels by modulating nedd (neural precursor cell expressed developmentally downregulated protein) 4-1 ubiquitin ligase
Circ Genom Precis Med 2019; 12(9):407-420 PMID 6750970
30. Zhang K, Yuan G, **Werdich AA**, Zhao Y
Ibuprofen and diclofenac impair the cardiovascular development of zebrafish (*Danio rerio*) at low concentrations
Environ Pollut 2020; 258:113613 PMID in progress
31. Ghazizadeh Z, Kiviniemi TO, Olafsson S, Plotnick D, Beerens M, Zhang K, Gillon L, Steinbaugh M, Barrera V, Ho Sui S, **Werdich AA**, Kapur S, Eranti A, Gunn J, Jalkanen J, Airaksinen J, Kleber AG, Hollmen M, MacRae CA
A metastable atrial state underlies the primary genetic substrate for myl4 mutation-associated atrial fibrillation
Circulation 2020; 141(4):301-312 PMID in progress
32. Zhang K, Liang J, Brun NR, Zhao Y, **Werdich AA**
Rapid Zebrafish Behavioral Profiling Assay Accelerates the Identification of Environmental Neurodevelopmental Toxicants
Environ Sci Technol 2021; 55(3):1919-1929
33. De Jong HN, Dewey FE, Cordero P, Victorio RA, Kirillova A, Huang Y, Madhvani R, Seo K, **Werdich AA**, Lan F, Orcholski M, Liu WR, Erbilgin A, Wheeler MT, Chen R, Pan S, Kim YM, Bommakanti K, Marcou CA, Bos JM, Haddad F, Ackerman M, Vasani RS, MacRae C, Wu JC, de Jesus Perez V, Snyder M, Parikh VN, Ashley EA
Wnt Signaling Interactor WTIP (Wilms Tumor Interacting Protein) Underlies Novel Mechanism for Cardiac Hypertrophy
Circ Genom Precis Med 2022; 15(4):e003563
34. Truslow J, Spillane A, Lin H, Cyr K, Ullal A, Arnold E, Huang R, Rhodes L, Block J, Stark J, Kretlow J, Beatty A, **Werdich A**, Banker D, Bianchi M, Shapiro I, Villapando J, Ravindarn S, Mance I, Phillips A, Earl J, Deo RC, Desai SA, MacRae CA
Understanding activity and physiology at scale: The Apple Heart & Movement Study
Npg Digital Medicine 2024; 7(1):242 PMID 11387614
35. Feher B*, **Werdich AA***, Chen CY, Barrow J, Lee SJ, Palmer N, Feres M
*Authors contributed equally to this work
Estimating periodontal stability using computer vision
J Dent Res 2025; 104(7):725-733
36. Feher B, de Souza Oliveira EH, Mendes Duarte P, **Werdich AA**, Giannobile WV, Feres M
Machine learning-assisted prediction of clinical responses to periodontal treatment
J Periodontol 2025; 96(11):1199-1212 PMID 12671691

Thesis:

1. Werdich AA

A laser ion source for an electron beam ion trap

Albert-Ludwigs-Universität Freiburg, Freiburg, Germany, Diplomarbeit (2000)

2. Werdich AA

A Nanophysiometer to study force – excitation coupling in single cardiac myocytes

Electronic dissertation, Vanderbilt University (2006) [http://etd.library.vanderbilt.edu/]

Manuscripts in preparation or submitted to preprint servers

1. Goto S, Homilius M, John JE, Truslow JG, **Werdich AA**, Blood AJ, Park BH, MacRae CA, Deo RC
Artificial intelligence-enabled event adjudication: estimating delayed cardiovascular effects of respiratory viruses. <https://www.medrxiv.org/content/10.1101/2020.11.12.20230706v1>

2. Goto S, **Werdich AA**, Homilius M, John JE, Gan LM, MacRae CA, DiCarli MF, Deo RC
Discovery of cardiac imaging biomarkers by training neural network models across diagnostic modalities
<https://www.medrxiv.org/content/10.1101/2021.02.07.21251025v1>

3. Csályi KD, Rharass T, Schulz M, Phan MHQ, Wakula P, Mhatre KN, Plotnick D, **Werdich AA**, Zauber H, Sury M, Selbach M, Heinzl FR, Klußmann E, Panáková D
Wnt11/Fzd7 signaling compartmentalizes AKAP2/PKA to regulate L-type Ca²⁺ channel
<https://www.biorxiv.org/content/10.1101/741637v1>

4. Patel K, Fletcher RR, Cubaka VK, Kateera F, Niyigena A, Estrada EH, Nkurunziza J, Bikorimana L, Jiwa A, Kubwimana M, Boatin AA, Hedt-Gauthier B*, **Werdich AA***

*Authors contributed equally to this work

Artificial intelligence to integrate multi-day images for enhanced detection of surgical site infections in rural Rwanda. *Submitted for publication*

5. Busch CV, **Werdich AA**, Tavelli L, Feres M, Feher B

Interpreting high-frequency dental ultrasonography using computer vision

Submitted for publication

Narrative Report

I am a physicist and biomedical engineer with research experience in human physiology, molecular biology, and data science. As the Director of Data Science and AI at the Core for Computational Biomedicine in the Department of Biomedical Informatics, I am working with a multidisciplinary team of scientists and engineers to develop innovative analytical and generative AI solutions that support education, research, and administration at HMS.

I earned my PhD in physics from Vanderbilt University and completed three years of postdoctoral training in cardiac electrophysiology and high-speed fluorescence imaging at Massachusetts General Hospital. I then continued my postdoctoral training at Case Western Reserve University, focusing on integrative cardiac electrophysiology and image analysis. During this time, I developed algorithms to visualize and quantify the propagation of action potentials and calcium concentrations across the heart's surface. This work is critical for understanding the mechanisms underlying life-threatening arrhythmia. My methods have been validated through multiple publications in high-impact journals. To date, I have published 36 original peer-reviewed papers in engineering and science journals, including five papers in Nature journals. I also developed computer vision algorithms and time-series analysis methods for the One Brave Idea project and the Apple Heart and Movement Study at Brigham and Women's Hospital. These projects aimed to identify novel predictors of early-onset coronary artery disease and explore relationships between physical activity and heart health using clinical and consumer data.

I developed strong grant-writing skills through successful funding applications. During my postdoctoral training, I received an NIH T32 Training Grant and an American Heart Association postdoctoral fellowship. I

was later awarded an NIH K99/R00 Pathway to Independence Award, which scored in the outstanding range. In 2015, I transitioned to the R00 phase and was promoted to Assistant Professor at Brigham and Women's Hospital. In 2024, I became a co-investigator with collaborators from the Harvard School of Dental Medicine on the HMS Dean's Innovation Award for the Use of Artificial Intelligence in Education, Research, and Administration. My team and I are currently contributing to several additional projects resulting from this initiative.

I am enthusiastic about continuing my work at the Core for Computational Biomedicine and contributing to innovation, discovery, and education at HMS. I plan to leverage my background in biomedical research and data science to build and grow a team of skilled data scientists and engineers. Together, we will develop shared data and analytical resources and provide expertise and project leadership for collaborative efforts with HMS investigators and educators.